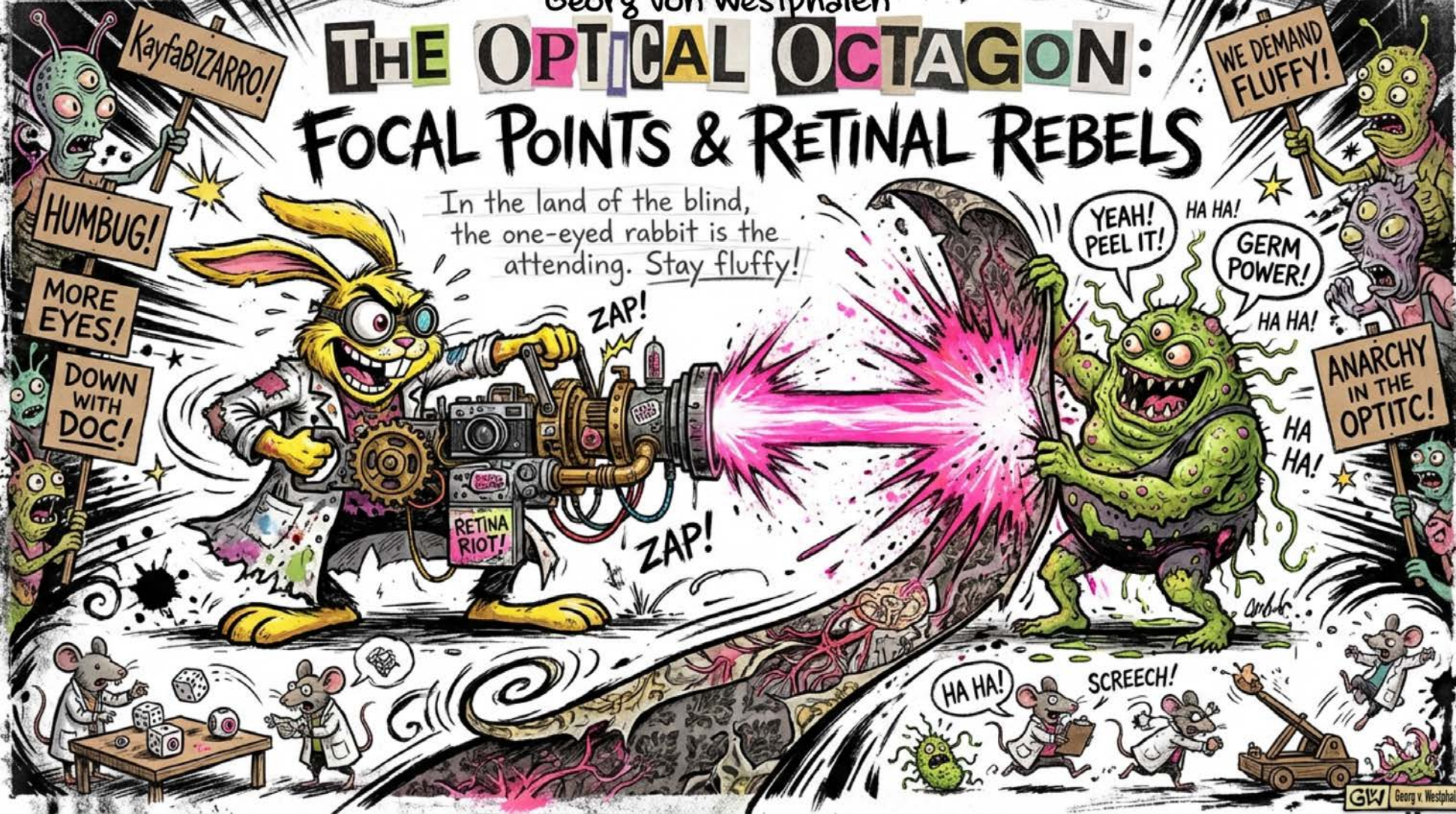


Georg von Westphalen

THE OPTICAL OCTAGON: FOCAL POINTS & RETINAL REBELS

In the land of the blind,
the one-eyed rabbit is the
attending. Stay fluffy!



Corny Cornea (The Window)

STAT: 40-43 Diopters
5 Layers
Avascular



POWER: The Clear Breach. If Corny is scratched, the Light Punks scatter into painful chaos.
LORE: The strongest refractor. Corny breathes O₂ directly from the air. Inflammation (Keratitis) creates permanent vandalism (scars) on the lens.



Aquas the Humor (The Pressure)

STAT: IOP 10-21 mmHg
Produced by Ciliary Body



POWER: The Inflation Pump. Fills the chamber to keep the eye from collapsing.
LORE: Drained via the Canal of Schlemm. If the drain is sabotaged, Aquas builds up until the pressure turns lethal for the nerves.

MedKayfabe!

Iris the Aperture (The Gate)

STAT: Mydriasis (Dilation)
vs. Miosis (Constriction)



POWER: The Light Gate. Regulates the photon-count entering the ring.
LORE: Controlled by Autonomic Tug-of-War. Sympathetics pull her wide (Fear); Parasympathetics shut her tight (Rest).



MedKayfabe!

Ciliary Sid (The Focuser)

STAT: Accommodation/
Zonular Tension



POWER: The Lens Squeezer. Sid pulls the strings to make the lens round for near-vision.
LORE: When Sid relaxes, the lens flattens for distance. As we age, the lens gets too stiff for Sid to squeeze (Presbyopia).

Crystal the Lens (The Refractor)

15-20 Diopters
No vessels



POWER: The Fine-Tuner. Provides the final adjustment for a sharp image on the sensor.
LORE: Crystal stays clear until proteins denature. She is the only part of the camera that can change shape.



What the
FLUFF?!



Cataract Carl (The Cloud)

Nuclear Sclerosis
Red Reflex ↓



POWER: The Fog Machine. Dims the world and creates a Halo glare.
LORE: Carl is the physical manifestation of oxidized lens proteins. He effectively pulls a dirty curtain over the camera.

MedKayfabe!

Phaco-Phil (The Shredder)

Ultrasound
Fragmentation



POWER: The Phaco-Blast. Phil pulverizes Cataract Carl in seconds and vacuums up the debris.
LORE: The modern surgical exorcism of the lens. Phil clears the stage for the artificial replacement.



What the
FLUFF?!



MedKayfabe!

IOL Irene (The Implant)

Artificial Power
Monofocal vs. Multifocal



POWER: The Permanent View. A plastic lens that restores clarity forever.
LORE: Irene cannot accommodate like Crystal did, so the patient usually trades near-vision for distance clarity.

Minus-Max (Myopia)

Axis too long
Focal point in
front of Retina



POWER: The Distance Blur. Sees the ant on his knee, but the bus is a ghost.
LORE: The eye is too long for the lens's power. Correct with Minus (Concave) glasses to push the light back.

Plus-Paul (Hyperopia)

Axis too short
Focal point behind
Retina



POWER: The Near-Focus Strain. Paul has to work hard even to see the horizon.
LORE: The eye is too short. The light aims behind the Retina. Correct with Plus (Convex) glasses to pull the light forward.

$$M_{\text{nat}} = 1$$

$$g = \frac{h \cdot c^2}{2}$$

$$y' = c \cdot t \cdot s$$

BLÖDSINN!

Asti-Gusti (Astigmatism)

Cylindrical Error
Toric Surface



POWER: The Fun-House Mirror. Everything is distorted because the focus is uneven.
LORE: Corny Cornea is shaped like a rugby ball instead of a basketball. Light has two different focal lines.

Presby-Pete (The Old-Age Sight)

Lens Elasticity ↓



POWER: The Arm-Extender. Pete has to hold the newspaper in the next room to read it.
LORE: Not a refractive error, but a mechanical failure. Crystal the Lens has simply become too stiff for Ciliary Sid to squeeze.

$$\frac{dv}{dt} = \frac{e \cdot \ln}{11a}$$

$$\frac{dv}{dx} = \frac{u \cdot T}{2}$$

MedKayfabe!

GW

Georg v. Westphalen

Glauco-Gus (The Silent Thief)

Cup-to-Disc Ratio ↑ / Perimetry Scars



POWER: The Peripheral Fade. Gus steals the edges of the world until only a tunnel is left.
LORE: High IOP kills the axons at the optic disc. It is painless and permanent.

Prosta-Pete (Latanoprost)

Prostaglandin Analog



POWER: The Drain Opener. Increases the back-door (uveoscleral) outflow of Aquas.
LORE: Pete makes the drain wider. Side effects: Long lashes and turning blue eyes brown.

Angle-Al (The Emergency)

IOP > 60 mmHg / Hard-as-rock Bulbus



POWER: The Red-Eye Rock. Sudden agony, nausea, and Halos around lights.
LORE: The door (angle) to the drain slams shut. Aquas is trapped. The optic nerve starts dying instantly.

Timmy-the-Block (Timolol)

Beta-Blocker



POWER: The Pump-Stopper. Cuts the production of Aquas at the source.
LORE: Less fluid entering means less pressure. Careful: Timmy can sneak into the lungs and cause Bronchitis.

GW

Georg v. Westphalen

KayfaBINGO!

Roddy & Conny (The Sensors)

120m Rods / 6m Cones / Fovea centralis



POWER: The Pixel Army. Roddy sees grey shadows; Conny sees neon colors.
LORE: Roddy is the night-fighter (Periphery); Conny is the day-sniper (Macula).

Call Doc Frizzle!

Detach-Dan (The Wallpaper Ripper)

Rhegmatogenous Tear / Vitreous Collapse



POWER: The Falling Curtain. A dark shadow moves across the field of vision.
LORE: Fluid gets behind the Retina and peels it off. Without the blood supply underneath, the Retina dies.

Mac-the-Degenerate (The Center Thief)

Drusen (Dry) vs. Wet (Hemorrhagic)



POWER: The Central Hole. You can see the clock, but never the time.
LORE: Mac destroys the fovea. In the Wet form, Annie-VEGF floods the sensor with blood.

Diabetic Debbie (Retinopathy)

Cotton Wool Spots / Neovascularization



POWER: The Capillary Pop. Sugar-sludge causes micro-aneurysms and leaks.
LORE: Chronic ischemia forces Debbie to summon Annie-VEGF, who builds fragiie, bleeding

GW

Georg v. Westphalen

Optic-Otto (The Cable)

CN II / 1.2 million Axons



POWER: The High-Speed Blackout.

LORE: If Otto is inflamed (Optic Neuritis), vision fades and eyes hurt when they move. The classic MS sign.

Vitreo-Vinnie (The Jelly)

99% Water / Collagen fibrils



POWER: The Floater Parade.

LORE: As Vinnie shrinks with age, he pulls away from the wall. His shadows are the flies you see in the sky.

Papil-Pam (The Signal)

Bilateral Papilledema / Blind Spot ↑



POWER: The Pressure Signal.

LORE: Pam isn't an eye disease; she's a brain alarm. High intracranial pressure pushes the nerves into the eye.

Annie-VEGF (The Saboteur)

Vascular Endothelial Growth Factor



POWER: The Illegal Road Builder.

LORE: Annie builds vessels where they don't belong. They leak, they bleed, and they cause scar.

Diagnosis?!

Pinky Pete (Conjunctivitis)

Viral/Bacterial/Allergic



POWER: The Red Itch.

LORE: Inflammation of the outer skin. Bacterial Pete brings the pus; Viral Pete brings the water.

Uveitis Ursula

Cells in Flare / Ciliary Flush



POWER: The Internal Revolt.

LORE: The immune system attacks the iris. Ursula loves to hang out with Rheuma-Rick.

Stay fluffy!

Keratitis Karl

Corneal Ulcer / Dendritic (HSV)



POWER: The Cornea-Ripper.

LORE: Karl is deep and dangerous. He leaves scars on Corny that light can't pass through.

Endo Ed (Endophthalmitis)

Post-op emergency



POWER: The Nuclear Option.

LORE: Infection inside the eyeball. Ed can turn the eye into a bag of pus in 24 hours.

Stay fluffy!

Blepha Bill (Blepharitis)

Meibomian Gland Dysfunction



POWER: The Crusty Margin.

LORE: Chronic inflammation of the lid. Bill clogs the oil pipes, making the tears evaporate too fast.

Stye Stan (Hordeolum)

Staph infection



POWER: The Painful Bump.

LORE: An acute pustule at the lid edge. Stan is a localized riot in a gland.

Chalazion Chester

Chronic Granuloma



POWER: The Cold Lump.

LORE: A painless, hard nodule. Chester is the aftermath of a blocked oil gland that didn't clear.

Entropion Enno

Inward Lid Fold



POWER: The Lash-Scratch.

LORE: The lid turns in, and the lashes act like sandpaper on Corny Cornea.

KayfaBONGO!

Strabi Stevie (Strabismus)

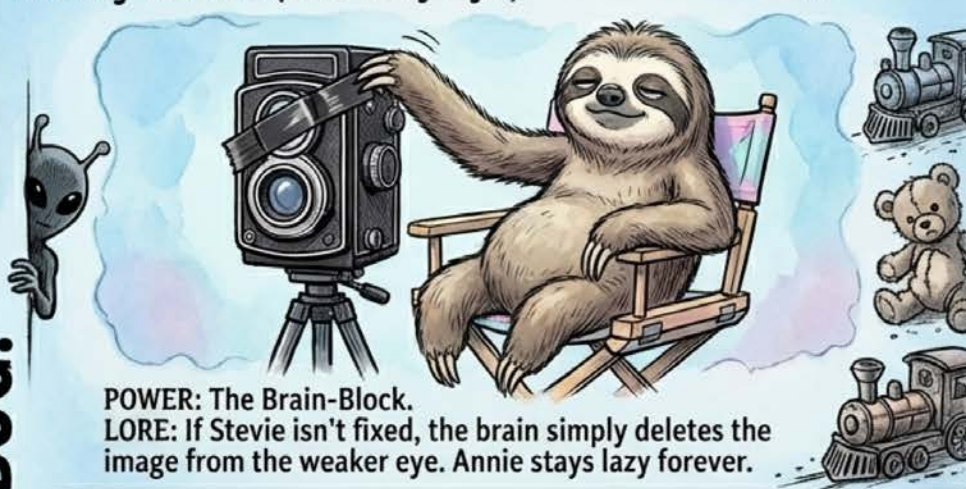
Tropia vs. Phoria

Ambly Annie (The Lazy Eye)

Suppression



POWER: The Misalignment.
LORE: The eyes aren't looking at the same spot. This creates two different stories for the brain.



POWER: The Brain-Block.
LORE: If Stevie isn't fixed, the brain simply deletes the image from the weaker eye. Annie stays lazy forever.

Leuko Leo (Leukocoria)

White Pupil / No Red Reflex

Obstruction Olli

Nasolacrimal Duct Block



POWER: The White Shadow.
LORE: The most dangerous sign in a child. Could be a Cataract or a Retinoblastoma monster.



POWER: The Constant Tear.
LORE: The drain to the nose is blocked. Olli makes the baby weep 24/7.

HUMBUG!

Horner Harry

Miosis / Ptosis / Anhidrosis



POWER: The Sympathetic Break.
LORE: The Fight wire is cut. The pupil stays small, the lid drops, and the face stays dry.

Argyll Robertson AI

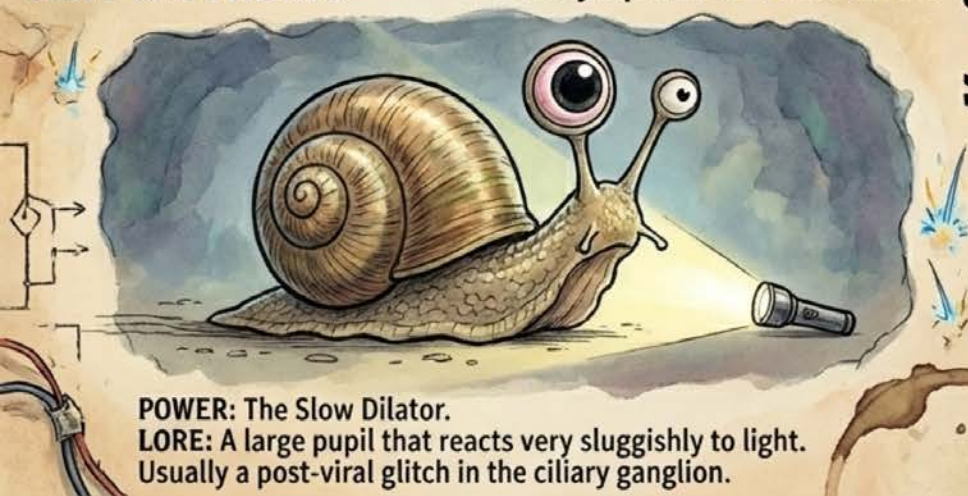
Light-Near Dissociation



POWER: The Prostitute's Pupil.
LORE: Reacts to near objects but ignores light. The classic red flag for Neurosyphilis.

Adie the Atonic

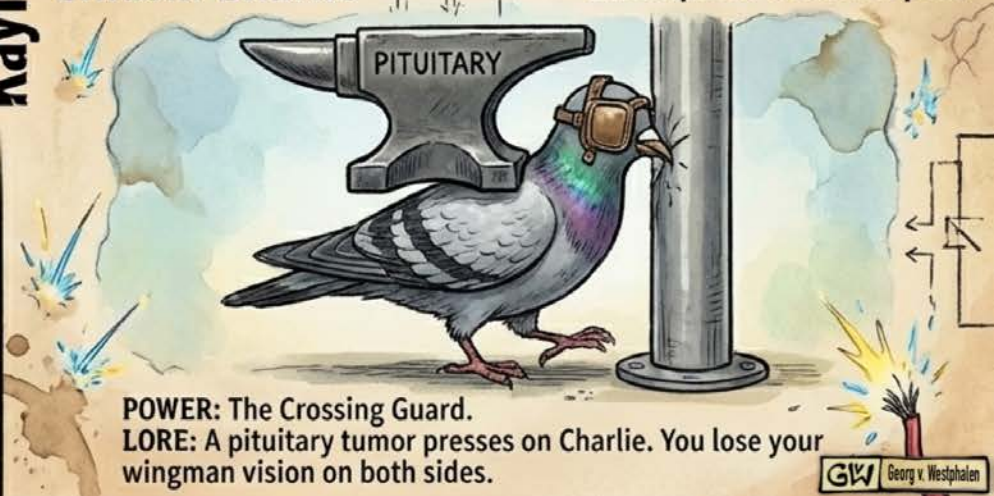
Parasympathetic Denervation



POWER: The Slow Dilator.
LORE: A large pupil that reacts very sluggishly to light. Usually a post-viral glitch in the ciliary ganglion.

Chiasm Charlie

Bitemporal Hemianopsia



POWER: The Crossing Guard.
LORE: A pituitary tumor presses on Charlie. You lose your wingman vision on both sides.

KayfaBONGO!

Scotoma Scott

Focal deficit



POWER: The Blind Spot.

LORE: A specific patch of the sensor is dead.

Hemi-Anopsia Hank

Stroke in Visual Cortex



POWER: The Half-Blind.

LORE: You lose the entire left or right side of the world.



KayfaBIZARRO!

Quadrant Quentin

Meyer's Loop Lesion



POWER: The Quarter-Thief.

LORE: Quentin only steals a slice of the pie (e.g., Pie in the Sky deficit).



Isopter Izzy

Perimetry Map



POWER: The Cartographer.

LORE: Izzy draws the map of where the sensors are working and where they are dark.





Atropine Alice
Antimuscarinic



POWER: The Permanent Wide-Eye.
LORE: Alice blocks the Constrict signal. You stay dilated for days and can't read.



Pilocarpine Phil
Muscarinic Agonist



POWER: The Drain Puller.
LORE: Phil forces the pupil to shrink, which pulls the trabecular meshwork open to drain Aquas.



MedKayfabe!



Dexa Dot
Topical Steroid



POWER: The Cooling Shield.
LORE: Stops the immune riot in Uveitis.
Warning: Long-term Dot invites Cataract-Carl.



Fluorescein Flo
Orange Dye / Blue Light



POWER: The Detective.
LORE: Flo stains the gaps in Corny Cornea. Under blue light, she reveals every ulcer and scratch.



Lasik Larry

Excimer Laser



POWER: The Cornea Carver.

LORE: Larry planes the surface of the eye to fix the focus.

Vitrectomy Vin

Posterior Segment Entry



POWER: The Jelly-Vac.

LORE: Vin removes Vinnie-the-Jelly to clear out blood or reach Detach-Dan.

Stay fluffy!

Buckle Bob

Scleral Encirclement

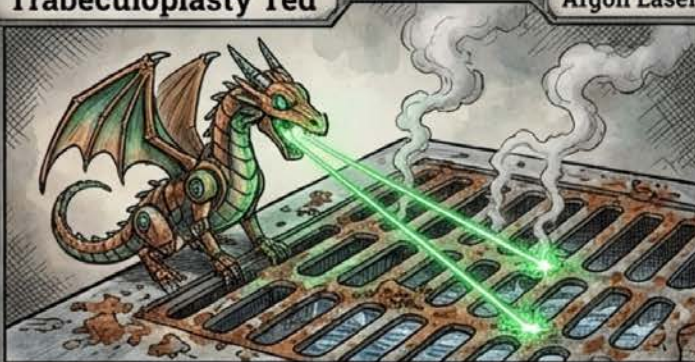


POWER: The Belt.

LORE: A silicone band squeezed around the eye to push the wall back against the peeling Retina.

Trabeculoplasty Ted

Argon Laser



POWER: The Drain Laser.

LORE: Ted burns small holes in the meshwork to help Aquas escape Gus-the-Glaucoma.



APOTHECARY

APOTHECARY

APOTHECARY

APOTHECARY

The Slit-Lamp SENTRY

Biomicroscopy



POWER: The Ultimate Inspection.
LORE: The Attending's favorite tool to find the Flare in the soup.

The Ophthalmoscope

Fundoscopy



POWER: The Deep Dive.
LORE: Looking at the only place in the body where you can see live nerves and vessels.

The Ophthalmology Attending

Boss Level



POWER: The Pimp-Master.
LORE: Localize the glitch. Is it the pump, the window, or the cable?

Stay fluffy! — and keep your eyes on the optics!

THE OPTICAL OCTAGON QUESTS

- Quest 1: The Red Rock.
(Identify Angle-AI)
- Quest 2: The Flying Curtain.
(Diagnose Detach-Dan)
- Quest 3: The Slow Thief.
(Identify Glauco-Gus)
- Quest 4: The Central Void.
(Diagnose Mac-the-Degenerate).